

AKASH SHRIVASTAVA

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PROFESSIONAL SUMMARY

Junior Research Fellow at NIT Raipur with expertise in Artificial Intelligence, Machine Learning, and Computer Vision. Experienced in designing and optimizing deep learning models for image dehazing, image deraining, and image restoration using TensorFlow, Python, and OpenCV. Passionate about developing efficient AI solutions for real-world computer vision and edge AI applications.

RESEARCH INTERESTS

Machine Learning • Deep Learning • Computer Vision • Image Enhancement • Image Dehazing • Artificial Intelligence

EDUCATION

National Institute of Technology (NIT), Raipur, Chhattisgarh, India 2023 – 2025
Master of Technology (M.Tech.) in Information Technology

Chhatrapati Shivaji Institute of Technology (CSIT), Durg, Chhattisgarh, India 2019 – 2023
Bachelor of Technology (Honours) in Computer Science and Engineering

PROFESSIONAL EXPERIENCE

National Institute of Technology (NIT), Raipur Raipur, Chhattisgarh, India
Junior Research Fellow (JRF) Jun 2025 – Present

- Working on the *Path Rakshak (Guardian of the Road)* project to enhance public transport safety through real-time driver and vehicle monitoring.
- Designed, implemented, and optimized deep learning architectures for single image dehazing and deraining using TensorFlow and Computer Vision techniques.
- Deployed Edge AI solutions on NVIDIA Jetson Nano for real-time, low-latency inference.
- Collaborated with researchers to improve the accuracy and efficiency of AI-based monitoring systems.

L&T Services Raipur, Chhattisgarh, India
Data Analyst Intern May 2021 – Mar 2022

- Managed databases and supported website development to improve system performance and reliability.
- Monitored L&T vehicles through an online tracking system, ensuring real-time status updates, data accuracy, and operational efficiency.
- Performed data analysis, reporting, and system optimization using analytical tools.

Vinayak Music World Raipur, Chhattisgarh, India
Computer Data Entry Operator Oct 2020 – Mar 2021

- Managed computer-based data entry operations while ensuring data accuracy and consistency.
- Installed and configured computer hardware, software, and user accounts.
- Monitored system performance and resolved basic technical issues to ensure smooth daily operations.

PROJECTS

Path Rakshak: AI-Based Road Safety Monitoring System [Ongoing]
TensorFlow, Python, OpenCV, NVIDIA Jetson Nano

- Developed and optimized deep learning models for image dehazing and image deraining to improve visual quality under adverse weather conditions.
- Designed lightweight deep learning models using TensorFlow for real-time image enhancement and visibility restoration.
- Deployed optimized AI models on NVIDIA Jetson Nano for efficient edge inference with low latency.
- Evaluated model performance using SSIM, PSNR, and other image quality metrics.

Enhancing Network Intrusion Detection Systems with Machine Learning



- Developed a machine learning-based Network Intrusion Detection System for real-time detection of malicious network activities.

- Applied feature selection and dimensionality reduction techniques to improve detection accuracy while reducing false positives.
- Evaluated and compared multiple machine learning algorithms using benchmark intrusion detection datasets.

Online Examination System



- Designed and developed a web-based examination management system for conducting online examinations.
- Implemented secure user authentication, exam scheduling, and automated result generation.
- Developed modules for student registration, question management, and performance evaluation.

PUBLICATIONS

Conference Paper

Published

- Akash Shrivastava *et al.*, “Enhancing Network Intrusion Detection Systems with Machine Learning: A Comparative Study with Intrusion Datasets,” *IEEE Conference Proceedings*, 2025.
- IEEE Xplore: <https://ieeexplore.ieee.org/abstract/document/10973753>

CERTIFICATIONS

- **Tools and Techniques for Information and Cyber Security (TICS-2024)**  Certificate
National Institute of Technology (NIT), Kurukshetra
- **Web Development Training**  Certificate
Internshala
- **22nd OITS International Conference on Information Technology**  Certificate
SRM University AP

AWARDS & RECOGNITIONS

- **Certificate of Appreciation** Certificate
Received for volunteering and contributing to the successful organization of the **13th IEEE International Conference on Intelligent Systems and Embedded Design (ISED-2025)**, organized by the Department of Computer Science & Engineering, National Institute of Technology (NIT) Raipur.
- **Fundamentals of Internet of Things (IoT)** Certificate
Successfully completed the **Fundamentals of Internet of Things (IoT)** training conducted by SAP Edutech.

SKILLS

- **Programming:** Python, SQL
- **Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Image Processing
- **Frameworks & Libraries:** TensorFlow, OpenCV, NumPy
- **Tools:** Git, VS Code, Jupyter Notebook, NVIDIA Jetson Nano
- **Soft Skills:** Problem Solving, Analytical Thinking, Research, Team Collaboration, Communication